

A Robust Real-Time Vision based GPS-denied Navigation System of UAV

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Abstract— In 2015 International UAV Innovation Grand Prix the competition, the cargo transport task is assumed as: there are 4 buckets placed in four circles on one moving platform. Firstly, the unmanned aerial vehicle (UAV) is required to identify circle targets and the black and white id marker near the circle on one moving platform, then the UAV chosen a target bucket, tracked and transported it to the other moving platform, until all 4 buckets are transported from one moving platform to the other. In order to accomplish the cargo transport task, a method of a real-time vision-based GPS-denied multiple object tracking for UAV is developed. The Pixhawk controller is used to achieve tracking, that the relative distance and velocity between the target and UAV is estimated by the image. Finally, the experimental results proved the effectiveness and robustness of the algorithm.

Keywords—UAV; Vision-based navigation; Target detection; Position and velocity estimation

I. INTRODUCTION

Vision-based object tracking system of UAV can be used in many situations such as debarking and landing of carrier aircraft[1]-[2] and cargo transportation[3]-[4]. Landing on a moving platform usually through recognition of the landmark or detection of the transport cargo' marker. Now days, some related research of Landing UAV in a moving platform or autonomous cargo transfer with UAV have come out. In such kind of research, GPS being used to localize and navigate outdoors cannot work well due to its low localization measurement accuracy, which can only attain meter level. In this case, vision is used to assist UAV to make more accurate location.

Appearance based visual-servoing method has been used in many occasions [5]. In this paper, the circle is chosen as landmark because it easy to be detected and the detection result is robust even there exists partial occlusions or blur. There are lots of researches on ellipse detection [6] - [8]. Comparing with the method of Hough transform [8], ellipse fitting [7] is a good way to find ellipses with given contours, but the ellipse fitting method cannot ensure the fitted ellipse is a correct detection. The affine moment invariants (AMIs) [9] is an effective way to detect the ellipse due to its all ellipses having equal AMIs.

In computer vision, a method called perspective-n-point (PnP)[10] is often used, in which 6 Degrees of freedom motion of camera from some known 2D-3D point correspondences is estimated, and it needs at least 4 points for calculation without ambiguities. Moreover there are some other camera position estimation methods like line correspondence or closed shape based method [11] [12]. These methods have a big drawback that is with a few pixels, a big position estimation error can have occurred. Hence position estimation based on circle landmark can greatly reduce the match error.

In this paper, we use a circle as the landmark [13], because it's simple shape, and the easy to be recognized by UAV in the sky. Moreover, we use the black and white square id marker [14] [15] to identify the circle. A robust real time algorithm including the detect and tracking of marker, position estimation and set point control of UAV is presented, because of the high frequency of the control updates which is needed to stabilize the UAV, an efficient algorithm is focused on the implementation.

This paper is organized as follow, Section 3 describes the related work about vision-based navigation of UAV, ellipse detection, pose estimation. The method of ellipse detection and black and white square id marker detection are described in section 4 and 5. In section 6 the method of position estimation is presented. The experimental results are presented in section 7 and the conclusion is given in section 8.

II. OVERVIEW OF THE VISION SYSTEM

A. Objectives

Our vision algorithms introduced in this paper can be used for detecting and tracking target by UAV in the situation of sea-rescue or cargo transportation. We did lots of experiments, and the experimental site is divided into two parts: the landing area and the target area. The landing area is used for UAV automatic take-off and landing. The target area, where some mobile target is located, is near the landing area, in which some mobile targets placed. Each of the targets is composed of a circle and the corresponding marker. The shape and the color of the targets are keys to the visual system to process. Vision system should be able to detect all the goals, and to intelligently choose a target that is used for

navigating the UAV. When the target moves, the visual system must give the real-time position and speed information of the target accurately.

B. Hardware design

The whole flight hardware system is constituted by four main parts: a bare rotor-craft platform, an on-board vision system, auto-pilot control system and a ground control system (GCS).

The on-board vision system includes: digital industrial camera, on-board computer, wireless adapter and power supply. The on-board computer for vision processing is an Advantech micro PC with the CPU as Intel Celeron J1900 (2x1.99 GHz). The processing speed of vision programs can reach 30fps in the on-board micro-computer. The structure of the system is shown in Fig. 1.

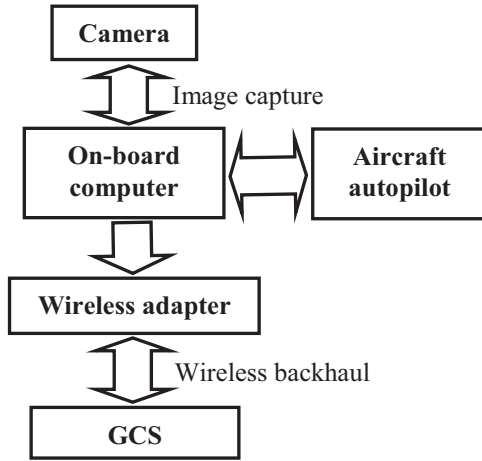


Fig.1 System Design

C. Working Procedure

The aircraft automatically takeoff in the landing area. Then the vision system starts operation and begin to process the images captured by the digital camera. When one or more circle be detected, the vision system calculates the relative height between the aircraft and the target area, as well as the relative three-dimensional position coordinates. The vision system guide aircraft to fly to the destination that over the target area about 10m height, and then start to detect the corresponding markers of each circle target. After all the markers are detected correctly, the vision system can select a circle intelligently according to its corresponding marker. Once the circle is selected, it will be tracked over the image sequence and its relative 3D-position and 3D-speed will be estimated continuously and accurately. To guide the UAV to locate and track the target, the vision system send the relative position and speed information to the aircraft autopilot. At last, the UAV put the processed images back to the ground control station, then we can see the real-time situation of the vision system.

From the above discussion, it is obvious that ellipse detection, marker detection, and single-circle-based position estimation are the core problems to be solved. Then we introduce our mainly proposed algorithms respectively.

III. ELLIPSE DETECTION

Ellipse detection algorithm is divided into two parts: whole ellipse detection and partial ellipse detection. In order to detect the ellipse, we need to get the contours in the image at first through the algorithm of edge detection. Once the contours are detected in the image, each of them will be processed by the following two steps to see if it corresponds to an ellipse.

A. Whole ellipse detection

Processing based on AMIs: This step is to compute the AMIs of a contour, then we compare the AMIs values with the theoretical values[16].

As proposed in [17], the three AMIs are expressed as

$$I_1 = (\mu_{20}\mu_{02} - \mu_{11}^2) / \mu_{00}^4, \quad (1)$$

$$I_2 = (\mu_{30}^2\mu_{03}^2 - 6\mu_{03}\mu_{30}\mu_{12}\mu_{21} + 4\mu_{30}\mu_{12}^3 + 4\mu_{21}^3\mu_{03} - 3\mu_{21}^2\mu_{12}^2) / \mu_{00}^{10}, \quad (2)$$

$$I_3 = ((\mu_{03}\mu_{21} - \mu_{12}^2)\mu_{20}) / \mu_{00}^7 + ((\mu_{21}\mu_{12} - \mu_{03}\mu_{30})\mu_{11} + (\mu_{12}\mu_{30} - \mu_{21}^2)\mu_{02}) / \mu_{00}^7, \quad (3)$$

And the moment μ_{ij} is given by

$$\mu_{ij} = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - \bar{x})^i (y - \bar{y})^j \rho(x, y) dx dy. \quad (4)$$

$(\bar{x}, \bar{y})$ is the coordinate of the centroid and $\rho(x, y)$ is the density distribution function. Suppose $\rho(x, y) = 1$ when (x, y) is on the circle, and $\rho(x, y) = 0$ otherwise[18]. Because all ellipses and circles have the same AMIs, about a circle, the parameters are set as: $\mu_{00} = 2\pi$, $\mu_{11} = 0$, $\mu_{02} = \mu_{20} = \pi$ and $\mu_{12} = \mu_{21} = \mu_{30} = \mu_{03} = 0$ [19]. Then we can get the theoretical values,

$$I_1 = \frac{1}{16\pi^2} \approx 0.00633, \quad (5)$$

According to our lots of experiments, we recommend the thresholds for I_1 , I_2 and I_3 to be ± 0.00025 , ± 0.0000002 and ± 0.000002 . As shown in Fig. 2, even in the presence of a large number of non-elliptical contours, the target that corresponding to ellipses are all successfully detected.

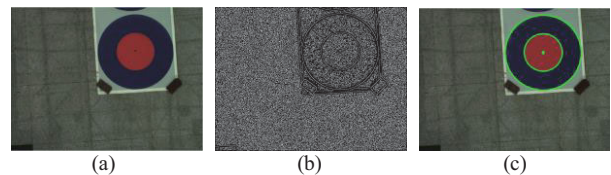


Fig. 2: An example to demonstrate the whole ellipse detection. (a) Color image; (b) Edge detection image; (c) Result of detection based on AMIs.

B. Partial Ellipse Detection

Actually, it is common that only parts of ellipse can be seen by digital camera due to the limited field of view. We next present an robust method to detect partial ellipses in the image obtained by the camera.

The method is executed by: computing the convex hull for each of the contours that are not classified as whole ellipses, fitting a ellipse for each convex hull, computing the algebraic error between the convex hull and the fitted ellipse. The algebraic error between the convex hull and its fitted ellipse is defined as follows. (u, v) is denoted as the coordinate of a pixel point in the image coordinate frame. And (u_c, v_c) denoted as the coordinate of the ellipse center. The a and b denote the lengths of the semi-major and semi-minor axes of the fitted ellipse respectively. Denote α as the angle between the major axis and the u -axis. The equation of the ellipse in the image frame is expressed as

$$\frac{[(u-u_c)\cos\theta + (v-v_c)\sin\theta]^2}{a^2} + \frac{[(u-u_c)\sin\theta - (v-v_c)\cos\theta]^2}{b^2} = 1, \quad (6)$$

The algebraic error is defined as

$$e_{\text{alg}} \triangleq \frac{1}{n} \sum_{i=1}^n \left| \frac{[(u_i-u_c)\cos\theta + (v_i-v_c)\sin\theta]^2}{a^2} + \frac{[(u_i-u_c)\sin\theta - (v_i-v_c)\cos\theta]^2}{b^2} - 1 \right|. \quad (7)$$

where (u_i, v_i) with $i = 1, \dots, n$ are the coordinates of the pixel points in the contour[20].

If the algebraic error is too large, the fitted ellipse of the contour will be classified as a false detection; otherwise, the contour can be classified as a partial ellipse. According to our lots of experiments, the threshold 0.05 can give a satisfactory result. As shown in Fig. 3, the partial ellipses are detected in the image.

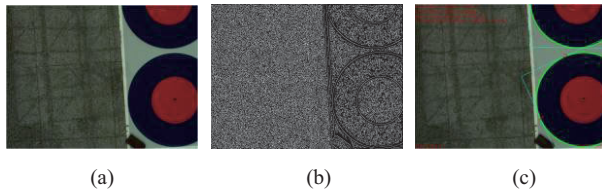


Fig. 3 An example to demonstrate the detection of partially occluded ellipses. The internal detected ellipses have been removed in the algorithm.

C. Summary of the Ellipse Detection Algorithm

The proposed ellipse detection algorithm is summarized as follows:

- 1) Filter color of the image, reserve the interested color, filter out the background color.
- 2) Detect contours in the image.

3) For each contour, compute its central moments, calculate I_1, I_2 and I_3 , and then compare the calculated I_1, I_2 and I_3 with the theoretical values. If

$$1) |I_1 - 1/(16\pi^2)| < 0.00025, \quad (8)$$

$$2) |I_2| < 0.0000002, \quad (9)$$

$$3) |I_3| < 0.000002, \quad (10)$$

the contour is a good candidate for a whole ellipse.

4) For each contour that does not satisfy (8),(9)and (10), get the convex hull of the contour, and fit an ellipse for each convex hull.

5) Compute its algebraic error between the fitted ellipse and the convex hull. If the error < 0.05, the contour is supposed a partial ellipse.

The ellipse detection algorithm could detect all the ellipses appeared in the image, while this algorithm could not choose which target can be used for navigating the UAV. We use the method of ID marker detection to choose a target as shown below.

IV. BLACK AND WHITE SQUIRE ID MARKER DETECTION

Multiple ellipses can be detected in one image, but we are only interested in one at a time. In order to find out which target to track, we need to give each target an id thus we can choose the one we want. So we need to use a maker that encoding the ID numbers, here we use a black and white squire ID marker with hamming code. In order to simplify the search procedure and make it faster, we use a marker with the size of 3x3 grids. It is shown as Fig. 4.



Fig.4. Black and write id marker

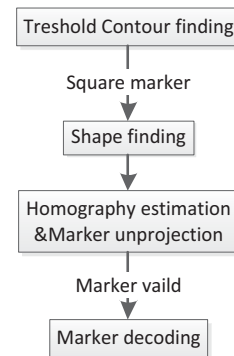


Fig.5. id marker detect process

The general procedure to identify id marker include several steps[21]. Firstly when we get a new frame, we should detect all the edges/borders in that frame, here we use the robust Canny Threshold method to do this work. Secondly, we need to convert border pixels to polygons. To obtain quadrangles candidates we choose the polygons which corner equals four, When several quadrangles have too short distances, we choose the one with longest circumference and get rid of the other then we

get the quadrangle candidates. Thirdly, we should estimate the homography and for each quadrangle in the list of candidates, unwarped the marker using homography to project it into a square and get the 2d code from the quadrangles. Last, we should decode the marker id from the 2d code. This procedure is shown as Fig.5.

We put the id maker near the circle. When we get the id of the maker, we can obtain the circle's id then choose the one we want. The detection result shown as Fig.6.

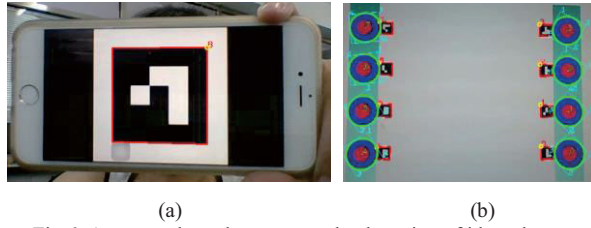


Fig.6. An example to demonstrate the detection of id marker

V. POSITION ESTIMATION

Position estimation will involve three coordinate frames, world frame, camera frame and image frame. Shown as Fig.7. Here we make world frame's center as the circle center and the Z-axis of world frame is orthogonal to the circle plane, so the center of the circle's coordinate in the world frame is $[0, 0, 0]$. All the points at that circle's Z-component should be zero. The origin of camera frame is the center of the camera, and the z-axis of the camera frame is original to the image plane. The origin of the image frame is the upper left corner of the image plane, and its u axis and v axis point rightwards and downwards respectively. This section we will introduce the way how to estimate the circle's relative position to the camera with the known four point correspondences, which is on the circle of world frame and their correspondent point on the image frame.

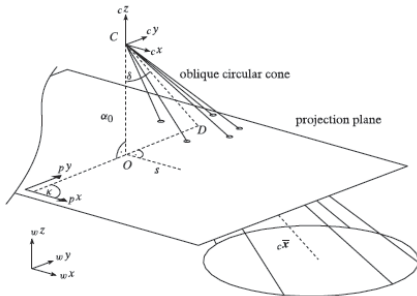


Fig.7. Perspective projection of a circle and the four correspondences

To be specific, we have a circle with known radius on the world frame, and we get the perspective projection of the circle which is an ellipse with known parameters in the image frame. And we know camera intrinsic matrix. With all this known conditions we need to calculate the coordinate of the circle in the camera frame. Thus we can get the distance between the circle and UAV.

We define $W = [X, Y, Z] \in R^3$ as the point in world frame, and $C = [x, y, z] \in R^3$ as the point in camera frame. The rotation vector and translation vector from world frame to camera frame is $R \in R^{3 \times 3}$, $T \in R^3$ respectively. Then we

have $C = RW + T$. We use subscript i to denote coordinate of center of the circle, the coordinate of the center of circle is $W_i = [0, 0, 0]$ in the world frame. We can get the coordinate of circle center is $C_i = RW_i + T$ in the camera frame. Suppose the image plane is parallel to the circle plane, then we have $C_i = T$.

In order to estimate T, we need at least four point correspondent from circle to ellipse. We have four vertex of the ellipse as n_1, n_2, n_3, n_4 , The parameter of ellipse is a, b, θ , Then the coordinates of four vertex are:

$$n_1 = (u_i + a \cos \alpha, v_i + a \sin \alpha) \quad (11)$$

$$n_2 = (u_i - b \sin \alpha, v_i + b \cos \alpha) \quad (12)$$

$$n_3 = (u_i - a \cos \alpha, v_i - a \sin \alpha) \quad (13)$$

$$n_4 = (u_i + b \sin \alpha, v_i - b \cos \alpha) \quad (14)$$

Suppose there are four point m_1, m_2, m_3, m_4 evenly distributed in one circle, then the line from point m_1 to m_3 (expressed as $m_1 m_3$) is orthogonal to $m_2 m_4$, so we can set the X and Y axis of world frame located in $m_1 m_3$ and $m_2 m_4$. If the radius of the circle is r, we can get $m_1 = [r, 0, 0]$, $m_2 = [0, -r, 0]$, $m_3 = [-r, 0, 0]$, $m_4 = [0, r, 0]$. Thus we get the four point correspondent $\{m_1, n_1\}$, $\{m_2, n_2\}$, $\{m_3, n_3\}$, $\{m_4, n_4\}$. Substitute the this point into position estimate algorithm[22], we can have the relative position between the circle and camera. The calculation method is shown as Fig.8. Here we use function solvePnP from OpenCV to solve T.

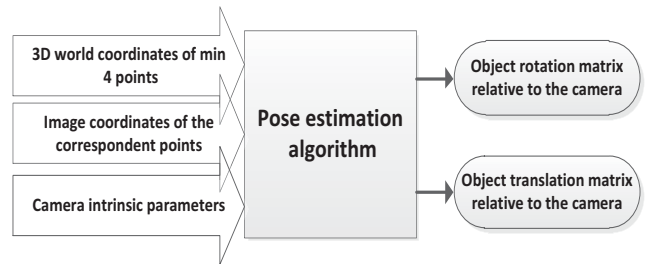


Fig.8. Pose estimation process



Fig.9. The indoor experiment setup.(a) Offline camera testing; (b)Indoor vision system testing based UAV

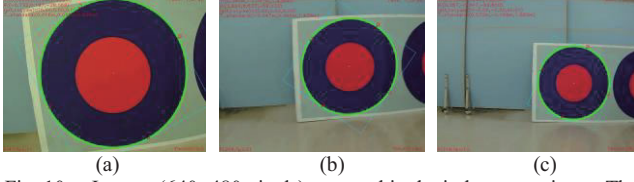


Fig. 10. Images (640×480 pixels) captured in the indoor experiment. The relative distance between Camera and target is (a)0.927m, (b)1.440m, (c)1.983m, respectively.

VI. EXPERIMENTAL RESULTS

A. Experimental results for position estimation

The indoor experimental setup is shown in Fig.9, and the outdoor experimental setup is shown in Fig.10. We compared the results measured by laser rangefinder with the results calculated by the vision system. We show a numerical experimental result to see if our algorithm can still work well in either indoor experiments (Fig.9) or outdoor experiments (Fig.11). The actual experimental results and comparative error are shown in TABLE I and TABLE II, respectively. The q_c in table is the position of the target center in the UAV coordinate system.



Fig.11. The outdoor experiment setup.(a)The unmanned helicopter platform;(b)The multi-rotor UAV platform.

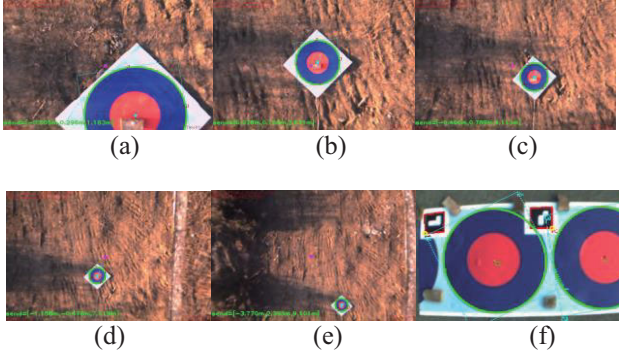


Fig.12. Images (640×480 pixels) captured in the outdoor experiment. The relative height of UAV is (a)1.189m, (b)2.627m, (c)4.263m, (d)7.176m, (e)10.054m, respectively. The image (f) is a integrated result of our target detection algorithms.

TABLE I. POSITION ESTIMATION RESULTS USING THE IMAGES IN FIG. 10 IN THE INDOOR EXPERIMENTS. (UNIT: M)

Image	Circle center q_c by vision(meter)	Range $ q_c $ by vision	Range $ q_c $ by laser rangefinder	Abs error of $ q_c $
(a)	(0.095,0.016,0.928)	0.931	0.927	0.002
(b)	(0.340,-0.047,1.439)	1.447	1.440	0.003
(c)	(0.570,0.164,1.883)	1.992	1.983	0.005

TABLE II. POSITION ESTIMATION RESULTS USING THE IMAGES IN FIG. 12 IN THE OUTDOOR EXPERIMENTS.(UNIT: M)

Image	Circle center q_c by vision	Range $ q_c $ by vision	Range $ q_c $ by laser rangefinder	Abs error of $ q_c $
(a)	(-0.505,0.296,1.183)	1.197	1.189	0.003
(b)	(0.078,0.155,2.631)	2.642	2.627	0.010
(c)	(-0.400,0.789,4.113)	4.284	4.263	0.019
(d)	(-1.158,-0.478,7.119)	7.228	7.176	0.038
(e)	(-3.770,2.393,9.101)	10.137	10.054	0.061

B. Results of Stability Analysis

In our indoor experiments, when the camera is placed statically, and the distance between the target and camera is about 5.8 m, the results of data stability are shown in Fig.13. The maximum fluctuation of location is about 0.002m, and the maximum fluctuation of speed is about 0.002m/s. In the outdoor experiments, the camera is installed on our UAV, and when the target move fast in different directions, the results of calculated position and speed are shown in Fig.14.

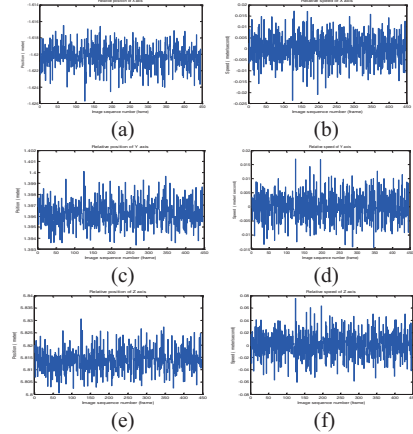


Fig. 13. The caculated results of stationary target .The pictures (a),(b),(c) represent the relative position of X,Y, and Z axis respectively in the UAV coordinate system . And (b),(d),(e) represent the relative speed of X axis, Y axis, and Z axis respectively.

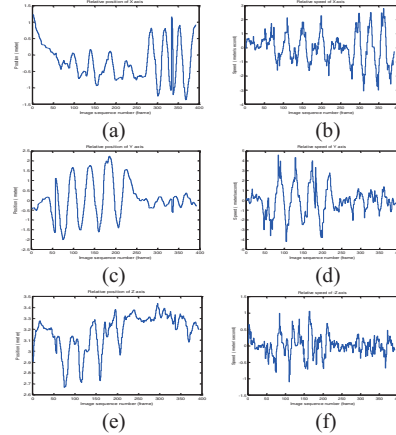


Fig.14. The caculated results of fast moving target. The pictures (a),(b),(c) represent the relative position of X, Y and Z axis respectively in the UAV coordinate system . And (b),(d),(e) represent the relative speed of X axis, Y axis, and -Z axis respectively.

TABLE III. AVERAGE TIME CONSUMPTION OF EACH PROCEDURE.

Procedure	Average Time Consumption
Undistortion	0.0157 sec
RGB to HSV	0.0030 sec
Color filter	0.0022 sec
Contour Detection	0.0011 sec
Ellipse Detection	0.0012 sec
Marker Detection	0.0014 sec
Position Estimation	0.0020 sec

The position and speed data of fast moving target proved the calculated result of the vision system is accurate and robust enough to navigate UAV to track the moving target. The vision system is more accurate than GPS when the target is within 10 meters, and it can be used for indoor navigation applications.

C. Time Consumption Test

The efficiency of the algorithms has been tested on the onboard computer with the CPU as Intel Celeron J1900 1.99GHz. The proposed algorithms are used to process 500 images captured by the digital camera. The size of each image is 320×240 pixels. The average time consumption of each procedure in our vision system is shown in TABLE III, each procedure consume little time.

As discussed above, based UAV we design a robust target detection method, and a accurate relative position and speed estimation method. The two methods are the core of the vision-based navigation system for UAV. The algorithms discussed above, like ellipse detection, marker detection and position estimation, are both robust and less time consuming, so they are very suitable for real-time applications.

VII. CONCLUSIONS AND FUTURE WORK

The experimental results shown that the vision system is to be as robust, efficient, and accurate. In the future, the robustness of the vision system can be improved through controlling the exposure time of the camera by the vision program automatically [23]. In addition, the algorithms of ellipse detection and marker detection can also be improved furtherly, such as combining with the machine learning and Hough transform method.

ACKNOWLEDGMENT

This work is supported by National Nature Sciences Foundation of China (Grant No. 61503369 and 61433016)

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